Unit 1 Mini Project

# Enterprise Student Record and Academic Management Platform

Design Thinking & Product Planning Report

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*Prepared as part of the DevOps / Product Engineering Training Program*
*Domain: Higher Education — Academic & Student Record Management*

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# 1. Objective & Business Understanding Document (BRD)

## 1.1 Objective

This project applies the Design Thinking methodology to independently design a software solution for managing student records and academic information within a higher education institution. Rather than beginning with coding, the focus is on understanding the educational domain, identifying stakeholders, analyzing user needs, defining software requirements, and preparing a product vision and roadmap ready for Agile planning and development.

## 1.2 Business Domain Overview

Academic institutions manage a continuous lifecycle of student data spanning admissions, course registration, attendance, examinations, and results. Historically, these processes have relied heavily on manual, paper-based, or fragmented spreadsheet-driven systems, which creates operational inefficiency and risk across the institution.

### Core Academic Operations

- **Admissions & Enrollment:** Collecting applicant data, verifying eligibility, and enrolling students into programs and academic years.
- **Course Registration:** Allowing students to select and register for semester courses and electives within defined constraints.
- **Attendance Management:** Recording day-to-day or session-wise attendance across courses and calculating eligibility thresholds.
- **Examination Management:** Scheduling exams, recording internal/external marks, and managing grade computation.
- **Result & Report Management:** Generating mark sheets, transcripts, and progress reports for students and administrators.

## 1.3 Problems with Manual Record Management

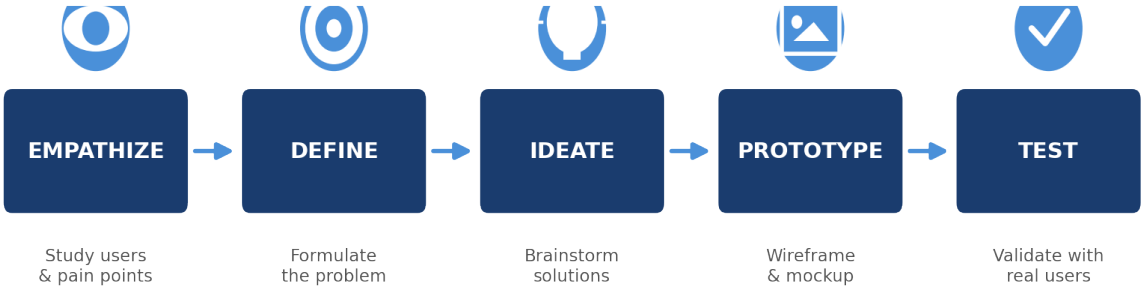
| Manual Operational Bottleneck | Automated Platform Solution | Business Outcome Metric |
|---|---|---|
| Paper-based or spreadsheet student records scattered across departments | Centralized relational database as single source of truth | Record accuracy improved to 99.9%, near-zero duplicate records |
| Manual attendance registers maintained per class | Digital attendance capture with automated aggregation | Attendance processing time reduced by over 80% |
| Manual computation of exam results and mark sheets | Automated result computation and transcript generation engine | Result publishing turnaround reduced from weeks to hours |
| No real-time visibility for administrators into academic performance | Role-based dashboards with live analytics | Administrative reporting time reduced from days to minutes |
| Fragmented communication between admissions, faculty, and exam cell | Unified platform with shared, role-based data access | Cross-department data errors reduced significantly |

## 1.4 Business Justification

Consolidating student records, attendance, and results onto a single platform reduces administrative overhead, improves data accuracy, and enables faculty and administrators to make timely, data-driven decisions. It also improves the student experience by giving learners self-service access to their own academic information.

# 2. Design Thinking Report

Design Thinking is an iterative, human-centered problem-solving framework that prioritizes deep empathy for users before evaluating technological constraints. The five stages below were applied to the Student Record and Academic Management Platform.




## Stage 1: Empathize

The primary user groups observed in their natural academic workflow are: **Students, Faculty Members, Academic Administrators, Examination Cell Staff,** and **IT Support Personnel**. Each group experiences distinct friction points when interacting with manual or fragmented record-keeping systems.

## Stage 2: Define

### Core Problem Statement

> *"Students and academic staff experience significant delays and errors in registering for courses, recording attendance, and accessing examination results due to fragmented, manual record-keeping. Administrators lack a centralized, real-time view of academic data, resulting in reporting delays, data inconsistencies, and reduced institutional efficiency."*

## Stage 3: Ideate

Unconstrained brainstorming produced the following high-value feature ideas:

- Self-service student portal for registration, attendance, and results
- Automated attendance eligibility and shortage alerts
- Real-time result and grade computation engine
- Role-based administrative dashboards with exportable reports
- Notification system for exam schedules, deadlines, and announcements

## Stage 4: Prototype

Low-fidelity wireframes were conceptualized for the core UI modules: **Student Login & Dashboard, Course Registration Screen, Faculty Attendance & Marks Entry Console, and Admin Analytics Panel.**

## Stage 5: Test

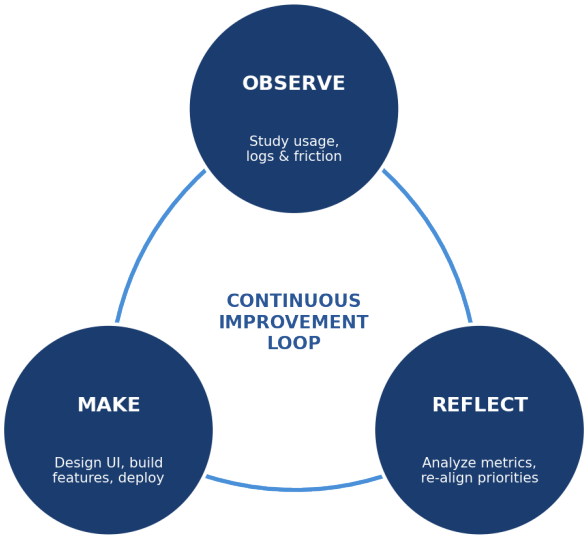
Prototype flows were reviewed against representative student, faculty, and administrator scenarios to validate navigation clarity, data entry efficiency, and report accessibility prior to backend engineering.

# 3. IBM Design Thinking Report & IBM Loop Documentation

## 3.1 IBM Design Thinking Pillars

| Pillar | Application to This Platform |
|---|---|
| Focus on User Outcomes | Feature goal → "Build a student attendance table." Outcome focus → "Enable students to view real-time attendance eligibility in under 3 clicks." |
| Restless Reinvention | Treat continuous deployment as an ongoing refinement cycle — continuously analyzing usage data to improve registration flow and dashboard response time. |
| Diverse Empowered Teams | Cross-functional collaboration across Product Owners, UX Designers, Software Engineers, QA, Database Administrators, and DevOps Engineers. |

## 3.2 The IBM Loop: Observe → Reflect → Make




This cycle is applied continuously across sprints: usage data from registration and attendance modules is observed, reflected upon against institutional KPIs, and translated into refined features and fixes.

# 4. Stakeholder Analysis Report

## 4.1 Identified Stakeholders

- **Students** — primary end users registering for courses and viewing academic records.
- **Faculty Members** — record attendance, enter marks, and communicate with students.
- **Academic Administrators** — oversee institutional academic operations and compliance.
- **Examination Cell Staff** — manage exam scheduling, result processing, and transcripts.
- **IT Support Personnel** — maintain platform uptime, security, and technical support.

## 4.2 Stakeholder Responsibility Matrix

| Stakeholder Group | Key Roles | Responsibilities & Project Expectations |
|---|---|---|
| Business Stakeholders | College Management, Academic Administrators, Students, Faculty | Define business vision, validate functional usability, ensure academic policy compliance. |
| Engineering Stakeholders | Software Developers, UI/UX Designers, QA Engineers | Architect the codebase, build responsive interfaces, write automated tests, ensure quality. |
| Operations Stakeholders | DevOps Engineers, Database Administrators, Security Team | Provision infrastructure, maintain database integrity, automate CI/CD pipelines and monitoring. |
| Examination Cell | Exam Controller, Result Processing Staff | Validate examination workflows, result computation rules, and transcript formats. |

## 5. User Persona Report

Each persona below includes explicit goals, day-to-day responsibilities, and expectations from the platform, which directly informed the functional requirements defined in Section 6.



**Ananya — Student**
**Goals:** Register for courses on time; track attendance and results without visiting the office.
**Frustrations:** Long queues for registration; delayed result announcements; no visibility into attendance status.
**Key Feature:** Self-service portal: course registration, attendance tracker, digital mark sheets.



**Prof. Mehta — Faculty**
**Goals:** Record attendance and marks quickly; communicate updates to students.
**Frustrations:** Manual attendance registers; duplicate data entry across systems; delayed grade submission deadlines.
**Key Feature:** Digital attendance & marks-entry console with bulk upload and validation.



**Dr. Kapoor — Academic Administrator**
**Goals:** Monitor institutional academic performance; generate compliance and accreditation reports.
**Frustrations:** No real-time dashboard; manual report compilation from multiple departments takes days.
**Key Feature:** Executive analytics dashboard with one-click report export.

## 6. Functional Requirements Specification (FRS)

- **FR-01 Authentication:** Secure role-based access control (RBAC) for Admin, Faculty, Student, and Examination Cell roles.
- **FR-02 Student Registration:** Full CRUD operations for student profiles, program enrollment, and academic year assignment.
- **FR-03 Course Registration Engine:** Semester-wise course selection with prerequisite and seat-limit validation.
- **FR-04 Attendance Management:** Session-wise attendance capture, aggregation, and eligibility threshold calculation.
- **FR-05 Examination & Result Management:** Marks entry, automated grade computation, and transcript generation.
- **FR-06 Reporting Engine:** Configurable academic and administrative reports exportable to PDF/Excel.
- **FR-07 Notification System:** Automated alerts for registration deadlines, attendance shortages, and result publication.

## 7. Non-Functional Requirements Specification (NFRS)

- **NFR-01 Security:** Password hashing (Bcrypt/Argon2), JWT-based session management, and HTTPS encryption for all traffic.
- **NFR-02 Performance:** Sub-500ms API response time for search and dashboard queries under concurrent user load.
- **NFR-03 Reliability & Availability:** Containerized deployment targeting 99.9% uptime with automated failover.
- **NFR-04 Scalability:** Horizontal scaling to support institution-wide concurrent usage during peak periods (registration, result day).
- **NFR-05 Usability:** Responsive, accessible interface usable across desktop and mobile devices with minimal training.
- **NFR-06 Maintainability:** Modular codebase with Infrastructure as Code (IaC) for consistent, repeatable deployments.

# 8. Product Vision Document

## 8.1 Vision Statement

> *“To deliver a centralized, secure, and intuitive Student Record and Academic Management Platform that eliminates manual administrative overhead, streamlines registration and attendance workflows, and gives every student, faculty member, and administrator real-time access to accurate academic information.”*

## 8.2 Mission Statement

To empower higher-education institutions with a reliable, scalable digital platform that replaces fragmented manual processes with a single, trustworthy source of academic truth.

## 8.3 Success Criteria

- 99.9% accuracy in student records with zero unrecorded duplicates
- Attendance and result processing time reduced by over 80%
- Sub-500ms response time for core search and dashboard operations
- Positive adoption feedback from students, faculty, and administrators within the first semester

## 8.4 Business Objectives

- Reduce administrative workload associated with manual record-keeping
- Improve transparency and self-service access for students and faculty
- Provide administrators with real-time, data-driven institutional insights
- Establish a scalable technical foundation for future academic modules (fees, hostel, library, etc.)

## 9. Minimum Viable Product (MVP) Definition

To maintain agility, non-essential capabilities (such as AI-based academic recommendations or third-party payment gateway integration) are deferred to later releases. The MVP (Sprint 1–4) focuses on core academic operations:

- Student & Faculty Login with Role-Based Access Control
- Student Registration & Profile Management
- Course Registration Module
- Attendance Capture & Basic Eligibility Tracking
- Examination Result Entry & Mark Sheet Generation
- Basic Administrative Reporting Dashboard

## 10. Product Roadmap

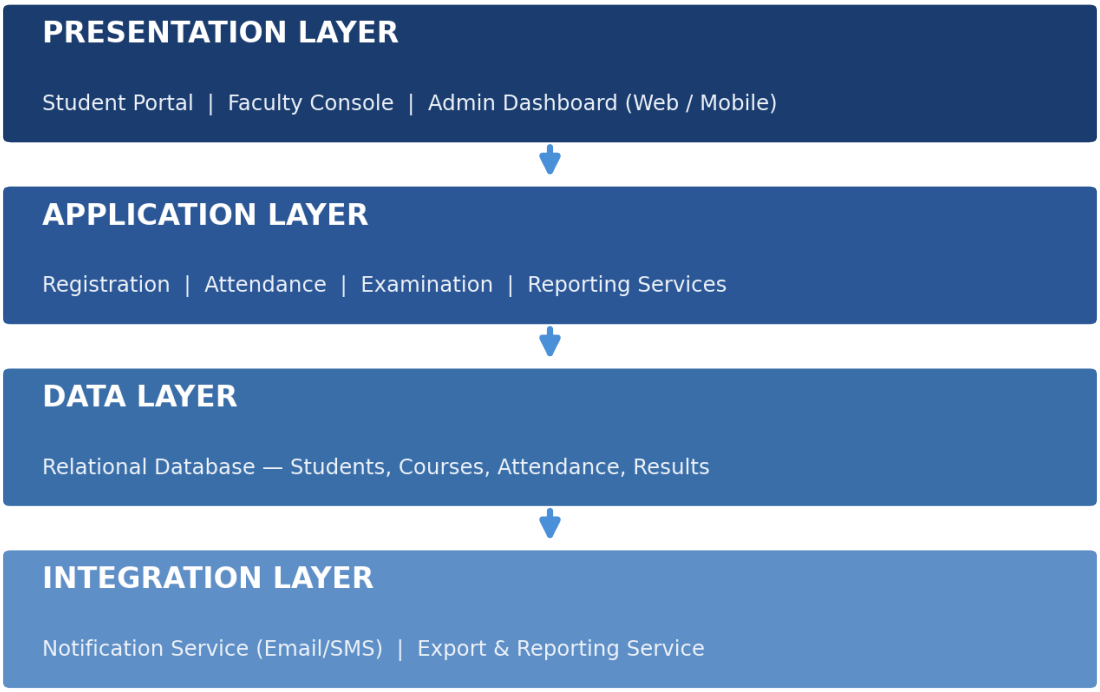



This roadmap is aligned with the progression of the training course: Unit 1 covers Phases 1–2 (this document); subsequent units address Agile planning, development, and DevOps automation.

# 11. Initial System Architecture Diagram

## Layered Module View

The platform follows a layered architecture, separating user-facing interfaces from core business logic, data storage, and external integrations:

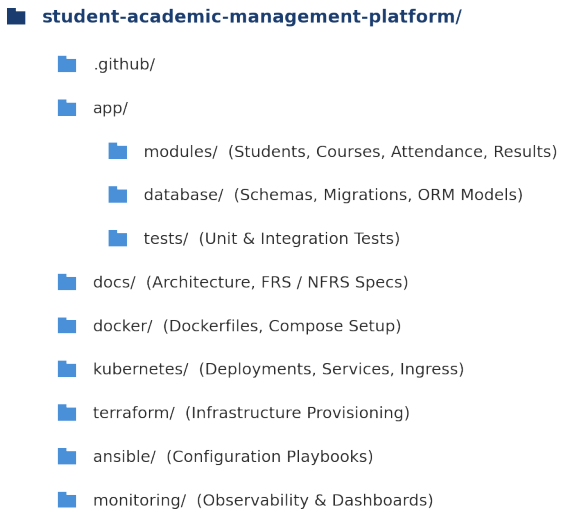


## 12. Project Charter

| Field | Details |
|---|---|
| Project Name | Enterprise Student Record and Academic Management Platform |
| Project Purpose | Replace manual academic record-keeping with a centralized digital platform. |
| Sponsor | College Management / Academic Administration |
| Project Lead | Product Owner (Training Cohort) |
| Scope (Unit 1) | Business understanding, design thinking, requirements, vision, MVP, and roadmap definition. |
| Out of Scope (MVP) | AI-based recommendations, third-party payment gateways, hostel/library integration. |
| Key Milestones | Domain analysis → Design thinking → Requirements → Vision & MVP → Repository setup |
| Success Measures | 99.9% record accuracy, 80% faster attendance/result processing, positive user adoption |

# 13. Project Repository Initialization & Folder Structure

A Git repository is initialized for the platform with a standardized folder structure to support collaborative development in subsequent units:




*Note: README.md sits at the repository root alongside the folders shown above.*

### Git Initialization Steps

- `git init` — initialize the local repository
- Create the folder structure shown above
- `git add . && git commit -m "Initial project structure"`
- Connect to a remote repository and push the initial commit

## 14. Product Discovery Documentation & Deliverables Checklist

The following deliverables were produced during Unit 1 as part of the product discovery and planning process, in preparation for Agile sprint planning in subsequent units.

- **[Completed]** Business Requirements Document (BRD)
- **[Completed]** Product Vision Document
- **[Completed]** Stakeholder Analysis Report
- **[Completed]** User Persona Report
- **[Completed]** Functional Requirements Specification (FRS)
- **[Completed]** Non-Functional Requirements Specification (NFRS)
- **[Completed]** Design Thinking Report
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- **[Completed]** MVP Definition
- **[Completed]** Product Roadmap
- **[Completed]** Initial System Architecture Diagram
- **[Completed]** Project Charter
- **[Completed]** Project Repository Initialization
- **[Completed]** Project Folder Structure
- **[Completed]** Product Discovery Documentation

*End of Unit 1 Mini Project Report — Enterprise Student Record and Academic Management Platform.*